

HW2 Handwritten Assignment

Yi-Lun Lee
b11901057@ntu.edu.tw

Che-Wei Tsai
f13945039@ntu.edu.tw

October 2025

Tools You Need to Know/Learn

- Backpropagation, Layer Normalization, Chain Rule
- Classification using GMM, Bayes' Rule
- AdaBoost, GradientBoost
- Bias–Variance Decomposition, Bregman Divergence, Kullback–Leibler Divergence
- Gradient Descent Convergence

You are expected to know (or learn through this homework) the definitions and usages of the above concepts, and are encouraged to discuss with TAs during TA hour if you encounter difficulties.

Homework Policy

- The homework is graded out of 100 points.
- The official submission deadline will follow the schedule announced on NTU COOL.
- Homework may be handwritten or typed (e.g. using $\text{\LaTeX}$), but must be submitted in **PDF format**.
- If you discuss the homework with classmates, you should state their student IDs in your submission.
- Plagiarism is strictly prohibited. Serious violations will be dealt with according to NTU regulations.

Problem 1 : (Layer Normalization) (20%)

Layer normalization is a popular normalization technique used in deep neural networks, particularly in **transformer** architectures. Unlike batch normalization which normalizes across the batch dimension, layer normalization normalizes across the feature dimension for each individual sample. The algorithm can be written as below:

Algorithm 1 Layer Normalization

Input Feature vector from a single data point $\mathbf{x} = (x_1, x_2, \dots, x_d)$ where d is the feature dimension

Output $\mathbf{y} = LN_{\gamma, \beta}(\mathbf{x})$

```

1: procedure LAYERNORMALIZE( $\mathbf{x}, \gamma, \beta$ )
2:    $\mu \leftarrow \frac{1}{d} \sum_{i=1}^d x_i$  ▷ layer mean
3:    $\sigma^2 \leftarrow \frac{1}{d} \sum_{i=1}^d (x_i - \mu)^2$  ▷ layer variance
4:   for  $i \leftarrow 1$  to  $d$  do
5:      $\hat{x}_i \leftarrow \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$  ▷ normalize
6:      $y_i \leftarrow \gamma \hat{x}_i + \beta$  ▷ scale and shift
7:   end for
8:   return
9: end procedure

```

During training we need to backpropagate the gradient of loss ℓ through this transformation, as well as compute the gradients with respect to the parameters γ, β .

- (a) (15pts) Towards this end, please write down the close form expressions for $\frac{\partial \ell}{\partial x_i}, \frac{\partial \ell}{\partial \gamma}, \frac{\partial \ell}{\partial \beta}$ in terms of $x_i, \mu, \sigma^2, \hat{x}_i, y_i$ (given by the forward pass) and $\frac{\partial \ell}{\partial y_i}$ (given by the backward pass).
 - Hint: You may first write down the close form expressions of $\frac{\partial \ell}{\partial \hat{x}_i}, \frac{\partial \ell}{\partial \sigma^2}, \frac{\partial \ell}{\partial \mu}$, and then use them to compute $\frac{\partial \ell}{\partial x_i}, \frac{\partial \ell}{\partial \gamma}, \frac{\partial \ell}{\partial \beta}$.
- (b) (5pts) Please explain why layer normalization needs learnable parameters γ and β to scale and shift after normalization.
 - Hint: What if different features should have different importance levels?

Solution:

- (a) We use the gradient descent to update γ and β :

$$\begin{aligned}\gamma &\leftarrow \gamma - \eta \frac{\partial \ell}{\partial \gamma} \\ \beta &\leftarrow \beta - \eta \frac{\partial \ell}{\partial \beta}\end{aligned}$$

where η is a learning rate and

$$\begin{aligned}\frac{\partial \ell}{\partial \gamma} &= \sum_{i=1}^d \left(\frac{\partial \ell}{\partial y_i} \frac{\partial y_i}{\partial \gamma} \right) = \sum_{i=1}^d \frac{\partial \ell}{\partial y_i} \hat{x}_i \\ \frac{\partial \ell}{\partial \beta} &= \sum_{i=1}^d \left(\frac{\partial \ell}{\partial y_i} \frac{\partial y_i}{\partial \beta} \right) = \sum_{i=1}^d \frac{\partial \ell}{\partial y_i}\end{aligned}$$

Note that we sum from 1 to d because layer normalization computes statistics across the feature dimension for each individual sample.

Now, we derive some important terms by chain rule:

$$\begin{aligned}\frac{\partial \ell}{\partial \hat{x}_i} &= \frac{\partial \ell}{\partial y_i} \frac{\partial y_i}{\partial \hat{x}_i} = \frac{\partial \ell}{\partial y_i} \gamma \\ \frac{\partial \ell}{\partial \sigma^2} &= \frac{\partial \ell}{\partial \hat{x}_i} \frac{\partial \hat{x}_i}{\partial \sigma^2} = -\frac{1}{2} \sum_{i=1}^d \frac{\partial \ell}{\partial \hat{x}_i} (x_i - \mu)(\sigma^2 + \epsilon)^{-\frac{3}{2}}\end{aligned}$$

$$\begin{aligned}
\frac{\partial l}{\partial \mu} &= \frac{\partial l}{\partial \hat{x}_i} \frac{\partial \hat{x}_i}{\partial \mu} \\
&= \sum_{i=1}^d \frac{\partial l}{\partial \hat{x}_i} \frac{\partial}{\partial \mu} (x_i - \mu)(\sigma^2 + \epsilon)^{-\frac{1}{2}} \\
&= \sum_{i=1}^d \frac{\partial l}{\partial \hat{x}_i} \left[\frac{\partial(x_i - \mu)}{\partial \mu} (\sigma^2 + \epsilon)^{-\frac{1}{2}} + (x_i - \mu) \frac{\partial(\sigma^2 + \epsilon)^{-\frac{1}{2}}}{\partial \mu} \right] \\
&= \sum_{i=1}^d \frac{\partial l}{\partial \hat{x}_i} \left[\frac{-1}{\sqrt{\sigma^2 + \epsilon}} + (x_i - \mu) \frac{\partial(\sigma^2 + \epsilon)^{-\frac{1}{2}}}{\partial \mu} \right]
\end{aligned}$$

We know $\sigma^2 = \frac{1}{d} \sum_{i=1}^d (x_i - \mu)^2$, so the second term in bracket is

$$\begin{aligned}
(x_i - \mu) \frac{\partial(\sigma^2 + \epsilon)^{-\frac{1}{2}}}{\partial \mu} &= (x_i - \mu) \frac{-1}{2} (\sigma^2 + \epsilon)^{-\frac{3}{2}} \frac{\partial(\sigma^2 + \epsilon)}{\partial \mu} \\
&= \frac{-1}{2} (x_i - \mu) (\sigma^2 + \epsilon)^{-\frac{3}{2}} \frac{\partial \left(\frac{1}{d} \sum_{i=1}^d (x_i - \mu)^2 + \epsilon \right)}{\partial \mu} \\
&= \frac{-1}{2} (x_i - \mu) (\sigma^2 + \epsilon)^{-\frac{3}{2}} \left(\frac{-2}{d} \sum_{i=1}^d (x_i - \mu) \right)
\end{aligned}$$

Hence,

$$\begin{aligned}
\frac{\partial l}{\partial \mu} &= \sum_{i=1}^d \frac{\partial l}{\partial \hat{x}_i} \frac{-1}{\sqrt{\sigma^2 + \epsilon}} + \underbrace{\sum_{i=1}^d \frac{\partial l}{\partial \hat{x}_i} \frac{-1}{2} (x_i - \mu) (\sigma^2 + \epsilon)^{-\frac{3}{2}} \left(\frac{-2}{d} \sum_{i=1}^d (x_i - \mu) \right)}_{\frac{\partial l}{\partial \sigma^2}} \\
&= \sum_{i=1}^d \frac{\partial l}{\partial \hat{x}_i} \frac{-1}{\sqrt{\sigma^2 + \epsilon}} + \frac{\partial l}{\partial \sigma^2} \left(\frac{-2}{d} \sum_{i=1}^d (x_i - \mu) \right)
\end{aligned}$$

To derive $\frac{\partial l}{\partial x_i}$, we use the chain rule $\frac{\partial l}{\partial x_i} = \frac{\partial l}{\partial \hat{x}_i} \frac{\partial \hat{x}_i}{\partial x_i} + \frac{\partial l}{\partial \sigma^2} \frac{\partial \sigma^2}{\partial x_i} + \frac{\partial l}{\partial \mu} \frac{\partial \mu}{\partial x_i}$. Now, calculate the remaining term:

$$\begin{aligned}
\frac{\partial \hat{x}_i}{\partial x_i} &= \frac{1}{\sqrt{\sigma^2 + \epsilon}} \\
\frac{\partial \mu}{\partial x_i} &= \frac{1}{d} \\
\frac{\partial \sigma^2}{\partial x_i} &= \frac{2(x_i - \mu)}{d}
\end{aligned}$$

That is,

$$\frac{\partial l}{\partial x_i} = \frac{\partial l}{\partial \hat{x}_i} \frac{1}{\sqrt{\sigma^2 + \epsilon}} + \frac{\partial l}{\partial \sigma^2} \frac{2(x_i - \mu)}{d} + \frac{1}{d} \frac{\partial l}{\partial \mu}$$

(b) The learnable parameters γ and β are necessary to restore the representational power of the network after normalization.

Problem 2 : (Classification with Gaussian Mixture Model) (20%)

In this question, we tackle the binary classification problem through the generative approach, where we assume the data point X (viewed as a $\mathbb{R}^d$ -valued r.v.) and its label Y (viewed as a $\{\mathcal{C}_1, \mathcal{C}_2\}$ -valued r.v.) are generated according to the generative model (parameterized by θ) as follows:

$$\mathbb{P}_\theta[X = \mathbf{x}, Y = \mathcal{C}_k] = \pi_k f_{\boldsymbol{\mu}_k, \Sigma_k}(\mathbf{x}) \quad (k \in \{1, 2\}) \quad (1)$$

where $\theta = (\pi_1, \pi_2, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma_1, \Sigma_2)$ for which

$$f_{\boldsymbol{\mu}_k, \Sigma_k}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}} \frac{1}{|\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)\right).$$

Now suppose we observe data points $\mathbf{x}_1, \dots, \mathbf{x}_N$ and their corresponding labels $y_1, \dots, y_N$, and $\pi_1 + \pi_2 = 1$.

- (a) (i) (2pts) Please write down the likelihood function $L(\theta)$ that describes how likely the generative model would generate the observed data $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ in terms of $\theta = (\pi_1, \pi_2, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma_1, \Sigma_2)$.
- (ii) (4pts) Find the maximum likelihood estimate $\theta^* = (\pi_1^*, \pi_2^*, \boldsymbol{\mu}_1^*, \boldsymbol{\mu}_2^*, \Sigma_1^*, \Sigma_2^*)$ that maximizes the likelihood function $L(\theta)$.
- (iii) (2pts) Write down $\mathbb{P}_\theta[Y = \mathcal{C}_1 | X = \mathbf{x}]$ and $\mathbb{P}_\theta[X = \mathbf{x} | Y = \mathcal{C}_1]$ in terms of $\theta = (\pi_1, \pi_2, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma_1, \Sigma_2)$. What are the physical meaning of the aforementioned quantities?
- (iv) (2pts) Express $\mathbb{P}_\theta[Y = \mathcal{C}_1 | X = \mathbf{x}]$ in the form of $\sigma(z)$, where $\sigma(\cdot)$ denotes the sigmoid function, and express z in terms of $\theta = (\pi_1, \pi_2, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma_1, \Sigma_2)$ and x .
- (b) (10pts) Suppose we pose an additional constraint that the covariance matrices of the two Gaussian distributions are identical, namely $\Sigma_1 = \Sigma_2 = \Sigma$, in which the generative model is parameterized by $\vartheta = (\pi_1, \pi_2, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma)$. Redo questions (a) under such setting.

Solution:

(a) (i) The likelihood function is given by

$$L(\theta) = \prod_{i=1}^N (1(y_i = \mathcal{C}_1) \pi_1 f_{\boldsymbol{\mu}_1, \Sigma_1}(\mathbf{x}_i) + 1(y_i = \mathcal{C}_2) \pi_2 f_{\boldsymbol{\mu}_2, \Sigma_2}(\mathbf{x}_i))$$

Since the indicator function is not differentiable, you may write it in another format.

W.L.O.G we may assume that there are N_1 numbers of $y_i \in \mathcal{C}_1$, N_2 numbers of $y_i \in \mathcal{C}_2$ and $N_1 + N_2 = N$. The likelihood function is given by

$$L(\theta) = \frac{1}{(2\pi)^{dN/2}} \pi_1^{N_1} \pi_2^{N_2} \frac{1}{|\Sigma_1|^{N_1/2}} \frac{1}{|\Sigma_2|^{N_2/2}} \times \prod_{i, y_i = \mathcal{C}_1}^{N_1} \exp\left(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu}_1)^T \Sigma_1^{-1}(\mathbf{x}_i - \boldsymbol{\mu}_1)\right) \prod_{j, y_j = \mathcal{C}_2}^{N_2} \exp\left(-\frac{1}{2}(\mathbf{x}_j - \boldsymbol{\mu}_2)^T \Sigma_2^{-1}(\mathbf{x}_j - \boldsymbol{\mu}_2)\right)$$

(ii) To make the calculation easier later, we change the above answer into log likelihood function

$$\begin{aligned} \log L(\theta) &= \log\left(\frac{1}{(2\pi)^{dN/2}}\right) + N_1 \log(\pi_1) + N_2 \log(\pi_2) + \frac{-N_1}{2} \log(|\Sigma_1|) + \frac{-N_2}{2} \log(|\Sigma_2|) + \\ &\quad \sum_{i, y_i = \mathcal{C}_1}^{N_1} \left(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu}_1)^T \Sigma_1^{-1}(\mathbf{x}_i - \boldsymbol{\mu}_1)\right) + \sum_{j, y_j = \mathcal{C}_2}^{N_2} \left(-\frac{1}{2}(\mathbf{x}_j - \boldsymbol{\mu}_2)^T \Sigma_2^{-1}(\mathbf{x}_j - \boldsymbol{\mu}_2)\right) \end{aligned}$$

Now we calculate the optimal π_1^*, π_2^* . Note that $\pi_1 + \pi_2 = 1$

$$\begin{aligned} \frac{\partial \log L(\theta)}{\partial \pi_1} &= \frac{N_1}{\pi_1} + \frac{N_2}{1 - \pi_1} = 0 \\ (1 - \pi_1)N_1 + \pi_1 N_2 &= 0 \Rightarrow \pi_1^* = \frac{N_1}{N} \end{aligned}$$

same for π_2 we have

$$\pi_2^* = \frac{N_2}{N}$$

for μ_1^*, μ_2^*

$$\begin{aligned} \frac{\partial \log L(\theta)}{\partial \mu_1} &= \sum_{i, y_i=C_1}^{N_1} (\Sigma_1^{-1}(\mathbf{x}_i - \boldsymbol{\mu}_1)) = \Sigma_1^{-1} \sum_{i, y_i=C_1}^{N_1} (\mathbf{x}_i - \boldsymbol{\mu}_1) = 0 \\ \sum_{i, y_i=C_1}^{N_1} (\mathbf{x}_i - \boldsymbol{\mu}_1) &= 0 \Rightarrow \mu_1^* = \frac{\sum_{i, y_i=C_1}^{N_1} x_i}{N_1} \end{aligned}$$

same for μ_2 we have

$$\mu_2^* = \frac{\sum_{i, y_i=C_2}^{N_2} x_i}{N_2}$$

for Σ_1^*, Σ_2^* , note that

$$\frac{\partial \log L(\theta)}{\partial \Sigma_1} = \frac{\partial \log L(\theta)}{\partial \Sigma_1^{-1}} \frac{\partial \Sigma_1^{-1}}{\partial \Sigma_1} = 0$$

Since the later one is not 0. We need the former one to be 0.

$$\begin{aligned} \frac{\partial \log L(\theta)}{\partial \Sigma_1^{-1}} &= \frac{1}{2} \frac{\partial -N_1 \log(|\Sigma_1|)}{\partial \Sigma_1^{-1}} + \frac{\partial \sum_{i, y_i=C_1}^{N_1} (-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu}_1)^T \Sigma_1^{-1}(\mathbf{x}_i - \boldsymbol{\mu}_1))}{\partial \Sigma_1^{-1}} \\ &= \frac{1}{2} \frac{\partial N_1 \log(|\Sigma_1^{-1}|)}{\partial \Sigma_1^{-1}} + -\frac{1}{2} \frac{\partial \sum_{i, y_i=C_1}^{N_1} \text{tr}((\mathbf{x}_i - \boldsymbol{\mu}_1)^T \Sigma_1^{-1}(\mathbf{x}_i - \boldsymbol{\mu}_1))}{\partial \Sigma_1^{-1}} \\ &= \frac{1}{2} \left(N_1 \Sigma_1^T - \frac{\partial \sum_{i, y_i=C_1}^{N_1} \text{tr}(\Sigma_1^{-1}(\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^T)}{\partial \Sigma_1^{-1}} \right) \\ &= \frac{1}{2} \left(N_1 \Sigma_1^T - \sum_{i, y_i=C_1}^{N_1} (\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^T \right) = 0 \\ \left(N_1 \Sigma_1^T - \sum_{i, y_i=C_1}^{N_1} (\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^T \right) &= 0 \Rightarrow \Sigma_1^* = \frac{\sum_{i, y_i=C_1}^{N_1} (\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^T}{N_1} \end{aligned}$$

same for Σ_2 we have

$$\Sigma_2^* = \frac{\sum_{i, y_i=C_2}^{N_2} (\mathbf{x}_i - \boldsymbol{\mu}_2)(\mathbf{x}_i - \boldsymbol{\mu}_2)^T}{N_2}$$

(iii) $\mathbb{P}_\theta[X = \mathbf{x} | Y = C_1] = \frac{\mathbb{P}(X=\mathbf{x}, Y=C_1)}{\mathbb{P}(Y=C_1)} = f_{\boldsymbol{\mu}_1, \Sigma_1}(\mathbf{x})$ which means the probability of x given the class

$\mathbb{P}_\theta[Y = C_1 | X = \mathbf{x}] = \frac{\mathbb{P}(X=\mathbf{x}, Y=C_1)}{\mathbb{P}(X=\mathbf{x})} = \frac{\pi_1 f_{\boldsymbol{\mu}_1, \Sigma_1}(\mathbf{x})}{\pi_1 f_{\boldsymbol{\mu}_1, \Sigma_1}(\mathbf{x}) + \pi_2 f_{\boldsymbol{\mu}_2, \Sigma_2}(\mathbf{x})}$ which means when we sample a new x how likely is it belongs to C_1

(iv) Same as the induction in class we have

$$z = \ln \frac{|\Sigma_2|^{1/2}}{|\Sigma_1|^{1/2}} - \frac{1}{2} x^T (\Sigma_1)^{-1} x + \mu_1^T (\Sigma_1)^{-1} x - \frac{1}{2} \mu_1^T (\Sigma_1)^{-1} \mu_1 + \frac{1}{2} x^T (\Sigma_2)^{-1} x - \mu_2^T (\Sigma_2)^{-1} x + \frac{1}{2} \mu_2^T (\Sigma_2)^{-1} \mu_2 + \ln \frac{N_1}{N_2}$$

(b) we only show those are modified.

(i)

$$\begin{aligned} L(\theta) &= \frac{1}{(2\pi)^{dN/2}} \pi_1^{N_1} \pi_2^{N_2} \frac{1}{|\Sigma|^{N/2}} \\ &\quad \prod_{i, y_i=C_1}^{N_1} \exp \left(-\frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu}_1)^T \Sigma^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_1) \right) \prod_{j, y_j=C_2}^{N_2} \exp \left(-\frac{1}{2} (\mathbf{x}_j - \boldsymbol{\mu}_2)^T \Sigma^{-1} (\mathbf{x}_j - \boldsymbol{\mu}_2) \right) \end{aligned}$$

(ii)

$$\Sigma^* = \frac{\sum_{i, y_i = C_1}^{N_1} (\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^T + \sum_{i, y_i = C_2}^{N_2} (\mathbf{x}_i - \boldsymbol{\mu}_2)(\mathbf{x}_i - \boldsymbol{\mu}_2)^T}{N}$$

(iii) the same

(iv)

$$z = (\mu_1 - \mu_2)^T \Sigma^{-1} x - \frac{1}{2} \mu_1^T (\Sigma)^{-1} \mu_1 + \frac{1}{2} \mu_2^T (\Sigma)^{-1} \mu_2 + \ln \frac{N_1}{N_2}$$

Problem 3 : (Multiclass AdaBoost) (20%)

Let $\mathcal{X}$ be the input space, $\mathcal{F}$ be a collection of multiclass classifiers that map from $\mathcal{X}$ to $[1, K]$, where K denotes the number of classes. Let $\{(x_i, \hat{y}_i)\}_{i=1}^m$ be the training data set, where $x_i \in \mathcal{X}$ and $\hat{y}_i \in [1, K]$. Given $T \in \mathbb{N}$, suppose we want to find functions

$$g_{T+1}^k(x) = \sum_{t=1}^T \alpha_t f_t^k(x), \quad k \in [1, K]$$

where $f_t \in \mathcal{F}$ and $\alpha_t \in \mathbb{R}$ for all $t \in [1, T]$. Here for $f \in \mathcal{F}$, we denote $f^k(x) = \mathbf{1}\{f(x) = k\}$, where $\mathbf{1}(\cdot)$ is an indicator function, as the k 'th element in the one-hot representation of $f(x) \in [1, K]$. The aggregated classifier $h : \mathcal{X} \rightarrow [1, K]$ is defined as

$$x \mapsto \operatorname{argmax}_{1 \leq k \leq K} g_{T+1}^k(x)$$

Please apply gradient boosting to show how the functions f_t and coefficients α_t are computed with an aim to minimize the following loss function

$$L((g_{T+1}^1, \dots, g_{T+1}^K)) = \sum_{i=1}^m \exp \left(\frac{1}{K-1} \sum_{k \neq \hat{y}_i} g_{T+1}^k(x_i) - g_{T+1}^{\hat{y}_i}(x_i) \right)$$

(10 pts for f_t and 10 pts for α_t , and derivation process is needed.)

Solution:

Given $\mathbf{g}_{t-1} = \{g_{t-1}^k\}_{k=1}^K$, we update $\mathbf{g}_t = \{g_{t-1}^k + \alpha_t f_t^k\}_{k=1}^K = \mathbf{g}_{t-1} + \alpha_t \mathbf{f}_t$ as follows:

$$\begin{aligned} \mathbf{f}_t &\in \operatorname{argmin}_{f \in \mathcal{F}} \frac{\partial}{\partial \alpha} L(\mathbf{g}_{t-1} + \alpha \mathbf{f}) \Big|_{\alpha=0} \quad (2pts) \\ &= \operatorname{argmin}_{f \in \mathcal{F}} \frac{\partial}{\partial \alpha} \sum_{i=1}^n \exp \left(\left(\frac{1}{K-1} \sum_{k \neq \hat{y}_i} g_{t-1}^k(x_i) - g_{t-1}^{\hat{y}_i}(x_i) \right) + \alpha \left(\frac{1}{K-1} \sum_{k \neq \hat{y}_i} f^k(x_i) - f^{\hat{y}_i}(x_i) \right) \right) \Big|_{\alpha=0} \quad (2pts) \\ &= \operatorname{argmin}_{f \in \mathcal{F}} \sum_{i=1}^n \exp \left(\frac{1}{K-1} \sum_{k \neq \hat{y}_i} g_{t-1}^k(x_i) - g_{t-1}^{\hat{y}_i}(x_i) \right) \left(\frac{1}{K-1} \sum_{k \neq \hat{y}_i} f^k(x_i) - f^{\hat{y}_i}(x_i) \right) \quad (3pts) \\ &= \operatorname{argmin}_{f \in \mathcal{F}} Z_t \mathbb{E}_{i \sim D_t} \left[\frac{1}{K-1} \sum_{k \neq \hat{y}_i} f^k(x_i) - f^{\hat{y}_i}(x_i) \right] \\ &= \operatorname{argmin}_{f \in \mathcal{F}} Z_t \mathbb{E}_{i \sim D_t} \left[\frac{1}{K-1} \cdot 1 \{f(x_i) \neq \hat{y}_i\} - 1 \{f(x_i) = \hat{y}_i\} \right] \\ &= \operatorname{argmin}_{f \in \mathcal{F}} Z_t \left(\frac{K}{K-1} \mathbb{P}_{i \sim D_t} [f(x_i) \neq \hat{y}_i] - 1 \right) = \operatorname{argmin}_{f \in \mathcal{F}} \mathbb{P}_{i \sim D_t} [f(x_i) \neq \hat{y}_i] \quad (3pts) \\ \alpha_t &\in \operatorname{argmin}_{\alpha \in \mathbb{R}} L(\mathbf{g}_{t-1} + \alpha \mathbf{f}_t) \quad (2pts) \\ &= \operatorname{argmin}_{\alpha \in \mathbb{R}} \sum_{i=1}^n \exp \left(\left(\frac{1}{K-1} \sum_{k \neq \hat{y}_i} g_{t-1}^k(x_i) - g_{t-1}^{\hat{y}_i}(x_i) \right) + \alpha \left(\frac{1}{K-1} \sum_{k \neq \hat{y}_i} f_t^k(x_i) - f_t^{\hat{y}_i}(x_i) \right) \right) \quad (3pts) \\ &= \operatorname{argmin}_{\alpha \in \mathbb{R}} Z_t \mathbb{E}_{i \sim D_t} \left[e^{\alpha \left(\frac{1}{K-1} \sum_{k \neq \hat{y}_i} f_t^k(x_i) - f_t^{\hat{y}_i}(x_i) \right)} \right] \\ &= \operatorname{argmin}_{\alpha \in \mathbb{R}} Z_t \mathbb{E}_{i \sim D_t} \left[e^{\frac{\alpha}{K-1}} \cdot 1 \{f_t(x_i) \neq \hat{y}_i\} + e^{-\alpha} \cdot 1 \{f_t(x_i) = \hat{y}_i\} \right] \\ &= \operatorname{argmin}_{\alpha \in \mathbb{R}} Z_t \left(\epsilon_t e^{\frac{\alpha}{K-1}} + e^{-\alpha} (1 - \epsilon_t) \right) = \left\{ \frac{K-1}{K} \log \frac{(K-1)(1-\epsilon_t)}{\epsilon_t} \right\} \quad (5pts) \end{aligned}$$

where

$$Z_t = \sum_{i=1}^n \exp \left(\frac{1}{K-1} \sum_{k \neq \hat{y}_i} g_{t-1}^k(x_i) - g_{t-1}^{\hat{y}_i}(x_i) \right)$$

and that D_t is a probability distribution for $t = 1, \dots, n$ given by

$$D_t(i) = \frac{1}{Z_t} \exp \left(\frac{1}{K-1} \sum_{k \neq \hat{y}_i} g_{t-1}^k(x_i) - g_{t-1}^{\hat{y}_i}(x_i) \right)$$

and that $\epsilon_t = \mathbb{P}_{i \sim D_t} [f_t(x_i) \neq \hat{y}_i]$ is the error of f_t on training sample weighted by the distribution D_t .

Problem 4 : (Bias–Variance Decomposition for Bregman Divergences) (20%)

Definition 1 (Strict Convexity). A differentiable function $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ is said to be strictly convex if for all distinct $p, q \in \mathbb{R}^d$ and all $\lambda \in (0, 1)$,

$$\phi(\lambda p + (1 - \lambda)q) < \lambda\phi(p) + (1 - \lambda)\phi(q).$$

Definition 2 (Bregman Divergence). The Bregman divergence associated with $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ is defined as

$$D_\phi(p||q) = \phi(p) - \phi(q) - \langle \nabla \phi(q), p - q \rangle,$$

for $p, q \in \mathbb{R}^d$.

- (a) (4pts) Prove that strict convexity implies uniqueness in the sense that

$$D_\phi(p||q) = 0 \iff p = q.$$

- (b) (6pts) Let $d = 1$. Suppose the true value is a fixed scalar $y \in \mathbb{R}$, and a learning algorithm produces a random prediction $\hat{Y}$ (due to training data or algorithm randomness). Denote the mean prediction by

$$\bar{Y} = \mathbb{E}[\hat{Y}].$$

Prove that

$$\mathbb{E}[D_\phi(y||\hat{Y})] = D_\phi(y||\bar{Y}) + \mathbb{E}[D_\phi(\bar{Y}||\hat{Y})].$$

is true if and only if $\langle \nabla \phi(\bar{Y}) - \mathbb{E}(\nabla \phi(\hat{Y})), y - \bar{Y} \rangle = 0$

- (c) (5pts) Let $d = 1$. Verify the decomposition for the special case $\phi(t) = f_1(t), t \in \mathbb{R}$, find $f_1(t)$ that can recover the classical MSE bias–variance identity

$$\mathbb{E}[(\hat{Y} - y)^2] = (\bar{Y} - y)^2 + \mathbb{E}[(\hat{Y} - \bar{Y})^2].$$

- (d) (5pts) Let $\phi(t) = f_2(t), t \in (0, \infty)^d$. Denote each component of the vectors $p = [p_1, p_2, \dots, p_d]^T$ and $q = [q_1, q_2, \dots, q_d]^T$. Show that we can properly select $\phi = f_2$ such that

$$D_\phi(p||q) = \sum_{i=1}^d \left(p_i \log \frac{p_i}{q_i} - p_i + q_i \right).$$

Furthermore, if p and q represent probability mass functions (so $\sum_{i=1}^d p_i = \sum_{i=1}^d q_i = 1$), show that in this case the Bregman divergence reduces to the Kullback–Leibler divergence:

$$D_\phi(p||q) = \sum_{i=1}^d p_i \log \frac{p_i}{q_i}.$$

(hint: For (c), you may use the result of (b))

Solution:

(a) $D_\phi(p\|q) = 0 \iff p = q$.
 $p = q \Rightarrow D_\phi(p\|q) = 0 : D_\phi(p\|p) = \phi(p) - \phi(p) - \langle \nabla \phi(p), p - p \rangle = 0$
 $D_\phi(p\|q) = 0 \Rightarrow D_\phi(p\|q) = 0 :$
 From strictly convex, $\phi(\lambda p + (1 - \lambda)q) < \lambda \phi(p) + (1 - \lambda)\phi(q)$.
 (b) For $d = 1$, recall the Bregman divergence:

$$D_\phi(p\|q) = \phi(p) - \phi(q) - \phi'(q)(p - q).$$

Compute the expectation:

$$\mathbb{E}[D_\phi(y\|\hat{Y})] = \mathbb{E}[\phi(y) - \phi(\hat{Y}) - \phi'(\hat{Y})(y - \hat{Y})].$$

Since $\phi(y)$ and y are constants:

$$\mathbb{E}[D_\phi(y\|\hat{Y})] = \phi(y) - \mathbb{E}[\phi(\hat{Y})] - y\mathbb{E}[\phi'(\hat{Y})] + \mathbb{E}[\hat{Y}\phi'(\hat{Y})]. \quad (1)$$

Now expand the right-hand side of the desired identity.

$$D_\phi(y\|\bar{Y}) = \phi(y) - \phi(\bar{Y}) - \phi'(\bar{Y})(y - \bar{Y}). \quad (2)$$

$$D_\phi(\bar{Y}\|\hat{Y}) = \phi(\bar{Y}) - \phi(\hat{Y}) - \phi'(\hat{Y})(\bar{Y} - \hat{Y}).$$

Taking expectation:

$$\mathbb{E}[D_\phi(\bar{Y}\|\hat{Y})] = \phi(\bar{Y}) - \mathbb{E}[\phi(\hat{Y})] - \bar{Y}\mathbb{E}[\phi'(\hat{Y})] + \mathbb{E}[\hat{Y}\phi'(\hat{Y})]. \quad (3)$$

$$D_\phi(y\|\bar{Y}) + \mathbb{E}[D_\phi(\bar{Y}\|\hat{Y})] = \phi(y) - \mathbb{E}[\phi(\hat{Y})] - y\mathbb{E}[\phi'(\hat{Y})] + \mathbb{E}[\hat{Y}\phi'(\hat{Y})] + \bar{Y}\mathbb{E}[\phi'(\hat{Y})]. \quad (4)$$

For the identity

$$\mathbb{E}[D_\phi(y\|\hat{Y})] = D_\phi(y\|\bar{Y}) + \mathbb{E}[D_\phi(\bar{Y}\|\hat{Y})]$$

to hold, the coefficients of y must match: From (1): coefficient of y is $-\mathbb{E}[\phi'(\hat{Y})]$. From (4): coefficient of y is $-\phi'(\bar{Y})$. Thus the identity holds iff:

$$\boxed{\mathbb{E}[\phi'(\hat{Y})] = \phi'(\bar{Y})}$$

Equivalently:

$$\boxed{\langle \phi'(\bar{Y}) - \mathbb{E}[\phi'(\hat{Y})], y - \bar{Y} \rangle = 0.}$$

This proves the desired “if and only if” condition.

(c) We want ϕ such that Bregman divergence reproduces the classical MSE identity:

$$\mathbb{E}[(\hat{Y} - y)^2] = (\bar{Y} - y)^2 + \mathbb{E}[(\hat{Y} - \bar{Y})^2].$$

Choose:

$$\boxed{\phi(t) = \frac{1}{2}t^2.}$$

Then: $\phi'(t) = t$, which is linear, put in to (b) $\langle \phi'(\bar{Y}) - \mathbb{E}[\phi'(\hat{Y})], y - \bar{Y} \rangle = 0$. Bregman divergence becomes:

$$D_\phi(p\|q) = \frac{1}{2}(p - q)^2.$$

Thus:

$$D_\phi(y\|\hat{Y}) = \frac{1}{2}(y - \hat{Y})^2, \quad D_\phi(y\|\bar{Y}) = \frac{1}{2}(y - \bar{Y})^2, \quad D_\phi(\bar{Y}\|\hat{Y}) = \frac{1}{2}(\hat{Y} - \bar{Y})^2.$$

Using the identity in part (b):

$$\mathbb{E}\left[\frac{1}{2}(y - \hat{Y})^2\right] = \frac{1}{2}(y - \bar{Y})^2 + \mathbb{E}\left[\frac{1}{2}(\hat{Y} - \bar{Y})^2\right].$$

Multiply by 2:

$$\mathbb{E}[(\hat{Y} - y)^2] = (y - \bar{Y})^2 + \mathbb{E}[(\hat{Y} - \bar{Y})^2].$$

So the correct f_1 is:

$$f_1(t) = \frac{1}{2}t^2.$$

(d) We want a ϕ such that

$$D_\phi(p||q) = \sum_{i=1}^d \left(p_i \log \frac{p_i}{q_i} - p_i + q_i \right).$$

Take a separable function:

$$\phi(p) = \sum_{i=1}^d f_2(p_i), \quad f_2(t) = t \log t.$$

Then:

$$f_2'(t) = \log t + 1.$$

Compute Bregman divergence:

$$D_\phi(p||q) = \sum_{i=1}^d [p_i \log p_i - q_i \log q_i - (\log q_i + 1)(p_i - q_i)].$$

Simplify:

$$= p_i \log \frac{p_i}{q_i} - p_i + q_i.$$

Thus:

$$D_\phi(p||q) = \sum_{i=1}^d \left(p_i \log \frac{p_i}{q_i} - p_i + q_i \right).$$

If p, q are pmfs so that $\sum_i p_i = \sum_i q_i = 1$:

$$D_\phi(p||q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

This is exactly the Kullback–Leibler divergence:

$$D_\phi(p||q) = \text{KL}(p||q).$$

Problem 5 (Gradient Descent Convergence) (2%)

Suppose the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable. Also, f is β -smoothness and α -strongly convex.

$$\beta - \text{smoothness} : \beta > 0, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|_2 \leq \beta \|\mathbf{x} - \mathbf{y}\|_2$$

$$\alpha - \text{strongly convex} : \alpha > 0, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, f(\mathbf{x}) - f(\mathbf{y}) - \nabla f(\mathbf{y})^T(\mathbf{x} - \mathbf{y}) \geq \frac{\alpha}{2} \|\mathbf{x} - \mathbf{y}\|_2^2$$

Then we propose a gradient descent algorithm

1. Find a initial $\boldsymbol{\theta}^0$.
2. Let $\boldsymbol{\theta}^{n+1} = \boldsymbol{\theta}^n - \eta \nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^n) \forall n \geq 0$, where $\eta = \frac{1}{\beta}$.

The following problems lead you to prove the gradient descent convergence.

1. Prove the property of β -smoothness function

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, f(\mathbf{x}) - f(\mathbf{y}) - \nabla f(\mathbf{y})^T(\mathbf{x} - \mathbf{y}) \leq \frac{\beta}{2} \|\mathbf{x} - \mathbf{y}\|_2^2$$

- (a) Define $g : \mathbb{R} \rightarrow \mathbb{R}, g(t) = f(\mathbf{y} + t(\mathbf{x} - \mathbf{y}))$. Show that $f(\mathbf{x}) - f(\mathbf{y}) = \int_0^1 g'(t) dt$.

Solution:

$$f(\mathbf{x} - \mathbf{y}) = g(1) - g(0) = \int_0^1 g'(t) dt.$$

- (b) Show that $g'(t) = \nabla f(\mathbf{y} + t(\mathbf{x} - \mathbf{y}))^T(\mathbf{x} - \mathbf{y})$.

Solution:

Let $\mathbf{z} = \mathbf{y} + t(\mathbf{x} - \mathbf{y})$.

$$g'(t) = \frac{dg(t)}{dt} = \frac{df(\mathbf{z})}{dt} = \sum_{i=1}^n \frac{df(\mathbf{z})}{dz_i} \frac{dz_i}{dt} = \sum_{i=1}^n \frac{df(\mathbf{z})}{dz_i} \frac{d(\mathbf{y}_i + t(\mathbf{x}_i - \mathbf{y}_i))}{dt} = \sum_{i=1}^n \frac{df(\mathbf{z})}{dz_i} (\mathbf{x}_i - \mathbf{y}_i) = \nabla f(\mathbf{z})^T(\mathbf{x} - \mathbf{y}) = \nabla f(\mathbf{y} + t(\mathbf{x} - \mathbf{y}))^T(\mathbf{x} - \mathbf{y})$$

- (c) Show that $|f(\mathbf{x}) - f(\mathbf{y}) - \nabla f(\mathbf{y})^T(\mathbf{x} - \mathbf{y})| \leq \int_0^1 |(\nabla f(\mathbf{y} + t(\mathbf{x} - \mathbf{y})) - \nabla f(\mathbf{y}))^T(\mathbf{x} - \mathbf{y})| dt$.

Solution:

$$\begin{aligned} |f(\mathbf{x}) - f(\mathbf{y}) - \nabla f(\mathbf{y})^T(\mathbf{x} - \mathbf{y})| &= \left| \int_0^1 g'(t) dt - \nabla f(\mathbf{y})^T(\mathbf{x} - \mathbf{y}) \right| \\ &= \left| \int_0^1 \nabla f(\mathbf{y} + t(\mathbf{x} - \mathbf{y}))^T(\mathbf{x} - \mathbf{y}) dt - \int_0^1 \nabla f(\mathbf{y})^T(\mathbf{x} - \mathbf{y}) dt \right| \\ &= \left| \int_0^1 (\nabla f(\mathbf{y} + t(\mathbf{x} - \mathbf{y}))^T - \nabla f(\mathbf{y})^T)(\mathbf{x} - \mathbf{y}) dt \right| \end{aligned}$$

- (d) By Cauchy-Schwarz inequality and the definition of β -smoothness, show that $|f(\mathbf{x}) - f(\mathbf{y}) - \nabla f(\mathbf{y})^T(\mathbf{x} - \mathbf{y})| \leq \frac{\beta}{2} \|\mathbf{x} - \mathbf{y}\|_2^2$, hence we get

$$f(\mathbf{x}) - f(\mathbf{y}) - \nabla f(\mathbf{y})^T(\mathbf{x} - \mathbf{y}) \leq \frac{\beta}{2} \|\mathbf{x} - \mathbf{y}\|_2^2$$

Solution:

By Cauchy-Schwarz inequality,

$$\begin{aligned} & \left| \int_0^1 (\nabla f(\mathbf{y} + t(\mathbf{x} - \mathbf{y}))^T - \nabla f(\mathbf{y})^T)(\mathbf{x} - \mathbf{y}) dt \right| \\ & \leq \int_0^1 |(\nabla f(\mathbf{y} + t(\mathbf{x} - \mathbf{y}))^T - \nabla f(\mathbf{y})^T)(\mathbf{x} - \mathbf{y})| dt \end{aligned}$$

$$\begin{aligned}
&\leq \int_0^1 |(\nabla f(\mathbf{y} + t(\mathbf{x} - \mathbf{y})) - \nabla f(\mathbf{y}))^T (\mathbf{x} - \mathbf{y})| dt \\
&\leq \int_0^1 \|(\nabla f(\mathbf{y} + t(\mathbf{x} - \mathbf{y})) - \nabla f(\mathbf{y}))\|_2 \|(\mathbf{x} - \mathbf{y})\|_2 dt \quad \text{By Cauchy-Schwarz inequality} \\
&\leq \int_0^1 \beta \|t(\mathbf{x} - \mathbf{y})\|_2 \|(\mathbf{x} - \mathbf{y})\|_2 dt \leq \int_0^1 t\beta \|(\mathbf{x} - \mathbf{y})\|_2^2 dt = \beta \|(\mathbf{x} - \mathbf{y})\|_2^2 \int_0^1 t dt = \frac{\beta}{2} \|(\mathbf{x} - \mathbf{y})\|_2^2
\end{aligned}$$

2. Let $\mathbf{y} = \mathbf{x} - \frac{1}{\beta} \nabla f(\mathbf{x})$ and use 1., prove that

$$f(\mathbf{x} - \frac{1}{\beta} \nabla f(\mathbf{x})) - f(\mathbf{x}) \leq -\frac{1}{2\beta} \|\nabla f(\mathbf{x})\|_2^2$$

and

$$f(\mathbf{x}^*) - f(\mathbf{x}) \leq -\frac{1}{2\beta} \|\nabla f(\mathbf{x})\|_2^2,$$

where $\mathbf{x}^* = \arg \min_{\mathbf{x}} f(\mathbf{x})$.

Solution:

By the fact that $f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \leq \frac{\beta}{2} \|\mathbf{x} - \mathbf{y}\|_2^2$,

$$f(\mathbf{x}) - f(\mathbf{x} - \frac{1}{\beta} \nabla f(\mathbf{x})) - \nabla f(\mathbf{x})^T \frac{1}{\beta} \nabla f(\mathbf{x}) \leq \frac{1}{2\beta} \|\nabla f(\mathbf{x})\|_2^2.$$

Also, $\nabla f(\mathbf{x})^T \frac{1}{\beta} \nabla f(\mathbf{x}) = \frac{1}{\beta} \|\nabla f(\mathbf{x})\|_2^2$. Hence,

$$f(\mathbf{x} - \frac{1}{\beta} \nabla f(\mathbf{x})) - f(\mathbf{x}) \leq -\frac{1}{2\beta} \|\nabla f(\mathbf{x})\|_2^2$$

Owing to the fact that $f(\mathbf{x}^*) \leq f(\mathbf{x}) - \frac{1}{\beta}$, we get

$$f(\mathbf{x}^*) - f(\mathbf{x}) \leq -\frac{1}{2\beta} \|\nabla f(\mathbf{x})\|_2^2.$$

3. Show that $\forall n \geq 0$,

$$\|\boldsymbol{\theta}^{n+1} - \boldsymbol{\theta}^*\|_2^2 = \|\boldsymbol{\theta}^n - \boldsymbol{\theta}^*\|_2^2 + \eta^2 \|\nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^n)\|_2^2 - 2\eta \nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^n)^T (\boldsymbol{\theta}^n - \boldsymbol{\theta}^*),$$

where $\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} f(\boldsymbol{\theta})$.

Solution:

$$\begin{aligned}
&\|\boldsymbol{\theta}^{n+1} - \boldsymbol{\theta}^*\|_2^2 = \|\boldsymbol{\theta}^n - \eta \nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^n) - \boldsymbol{\theta}^*\|_2^2 \\
&(\boldsymbol{\theta}^n - \eta \nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^n) - \boldsymbol{\theta}^*)^T (\boldsymbol{\theta}^n - \eta \nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^n) - \boldsymbol{\theta}^*) = \|\boldsymbol{\theta}^n - \boldsymbol{\theta}^*\|_2^2 + \eta^2 \|\nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^n)\|_2^2 - 2\eta \nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^n)^T (\boldsymbol{\theta}^n - \boldsymbol{\theta}^*)
\end{aligned}$$

4. Use 2. and the definition of α -strongly convex to prove $\forall n \geq 0$

$$\|\boldsymbol{\theta}^{n+1} - \boldsymbol{\theta}^*\|_2^2 \leq (1 - \frac{\alpha}{\beta}) \|\boldsymbol{\theta}^n - \boldsymbol{\theta}^*\|_2^2,$$

where $\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} f(\boldsymbol{\theta})$.

Solution:

By rearranging the inequality we get in 2., $\eta^2 \|\nabla f(\boldsymbol{\theta}^n)\|_2^2 \leq 2\beta \eta^2 (f(\boldsymbol{\theta}^*) - f(\boldsymbol{\theta}^n))$.

Also, by the definition of α -strongly convex, $-2\eta \nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}^n)^T (\boldsymbol{\theta}^n - \boldsymbol{\theta}^*) \leq -\alpha \eta \|\boldsymbol{\theta}^n - \boldsymbol{\theta}^*\|_2^2 - 2\eta (f(\boldsymbol{\theta}^*) - f(\boldsymbol{\theta}^n))$ Combined the above results and $\eta = \frac{1}{\beta}$, we get

$$\|\boldsymbol{\theta}^{n+1} - \boldsymbol{\theta}^*\|_2^2 \leq \|\boldsymbol{\theta}^n - \boldsymbol{\theta}^*\|_2^2 + \frac{2}{\beta} (f(\boldsymbol{\theta}^*) - f(\boldsymbol{\theta}^n)) - \frac{\alpha}{\beta} \|\boldsymbol{\theta}^n - \boldsymbol{\theta}^*\|_2^2 - \frac{2}{\beta} (f(\boldsymbol{\theta}^*) - f(\boldsymbol{\theta}^n)) = (1 - \frac{\alpha}{\beta}) \|\boldsymbol{\theta}^n - \boldsymbol{\theta}^*\|_2^2.$$

5. Use the above inequality to show that θ^n will converge to θ^* when n goes to infinity.

Solution:

Because $\frac{\alpha}{\beta} > 0$, also, by the property of β -smoothness function and the definition of α strongly convex function, we get $\alpha \leq \beta$, which derives $\frac{\alpha}{\beta} < 1$ and $0 < 1 - \frac{\alpha}{\beta} < 1$. When n goes to infinity, $\|\theta^n - \theta^*\| = 0$. Hence, θ^n will converge to θ^*